

Operational support systems — the global dimension

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Global activity in the telecommunications industry is increasing as operators look to expand into overseas markets, a trend that is currently being accelerated by the rationalisation of the information technology (IT), telecommunications and entertainment industries. This paper gives an overview of BT's global operational support systems (OSS) strategy and ambitions.

1. Introduction

There is increasing global activity in the telecommunications industry as operators look to expand into territories outside their home markets. This trend is currently being accelerated by the rationalisation of the information technology (IT), telecommunications and entertainment industries. This paper will give an overview of BT's global operational support systems (OSS) strategy which is a key enabler to supporting BT's global ambitions. It will describe the complex set of drivers that determine BT's global systems strategy. It concludes with an overview of the systems strategy, focusing on the evolving needs of BT's global ventures.

2. Globalisation of BT

2.1 The business drivers

BT's stated aim is to be 'the most successful worldwide telecommunications group'. BT's globalisation plans are coinciding with the rapid liberalisation of the world's telecommunications markets, with the expectation of second and third operator licences being issued in countries previously serviced by a single incumbent PTT. This is leading to a complex web of mergers, acquisitions and alliances spanning both regions and industry sectors.

BT is achieving its aim through a series of alliances, aiming for second operator licences in targeted countries, in partnership with local companies. These alliances are generically termed national partners. This is a role reversal to BT's operations in its UK home market. BT has also formed a global joint venture, Concert, in partnership with MCI, to provide a global services portfolio to corporate customers.

2.2 The technical challenges

The technical challenge for BT and its global partners is to deliver world-class global and domestic services consistently across the alliance platforms with the optimum combination of cost and quality of service. To achieve this, BT has defined, and is continuously evolving, a global technical strategy, driven by the business drivers, that encompasses both the networks and systems architecture. The aim of this strategy is to:

- define optimum target solutions towards which service platforms can migrate,
- enable BT's partners to deliver rapidly and economically new products and services, in advance of the incumbent PTT,
- avoid bespoke (high-cost, stove-pipe) implementations wherever possible,
- leverage commonality and hence achieve economies of scale.

In the increasingly competitive global market-place, where a lot of the network capabilities are becoming commoditised, differentiation between operators is being enabled more and more by sophisticated management capabilities. To specify and develop appropriate systems, the implications of these business drivers need to be analysed in more detail.

3. Systems implications of global business drivers

3.1 Distributed Management

The establishment of a number of alliances and distributorships across the globe, could lead to each alliance eventually ending up with its own stack of management systems. In addition, BT has developed its own regional management centres which, together with the Concert global management centre, form a comprehensive global management capability.

The systems within these centres need to operate in a multi-organisational business model, where end-to-end responsibility to deliver the global services is shared between these independent entities, although owned by only one. Thus management of the service is distributed between organisations. This places a prime driver on the systems interworking capabilities, as a strong systems coupling between centres will ensure that end-to-end management of the service is not compromised. This coupling is being defined in terms of a broker-interface architecture, where messages (processes and data), are brokered between centres and their relevant systems, as shown schematically in Fig 1.

3.2 Local autonomy versus global interworking

The product portfolios of BT's national partners usually include both Concert's global services and specific national services, tailored to the local market. The Concert services are targeted at the corporate customer segment, and are a key component of each national partner's product portfolio. These services come with their own set of defined global business processes and procedures, ensuring seamless interworking between national partners and sophisticated end-to-end management. The interorganisational procedures defining both the processes and the data associated with them, are encapsulated into the broker interface architecture. The process and data specifications will follow established and robust international standards wherever feasible. Each national partner will need to ensure that its systems comply with the broker interface.

The national product portfolio of each national partner focuses on local market drivers, which will inevitably vary from country to country, and region by region. To satisfy these local needs, national partners will, over time, evolve

systems architectures which meet their domestic product specifications and business processes. Thus a degree of local autonomy is vital to ensure that each national partner can react rapidly to its own market conditions.

Each national partner therefore has to integrate the predefined global requirements with its own domestic needs. This could potentially place conflicting drivers on the systems designs of national partners. BT is developing a fully integrated national partner systems and networks stack which by definition will comply with the broker-interface architecture. This will enable seamless global and domestic services. By choosing their systems and networks components from BT's global strategy, a national partner should minimise its whole-life systems and networks costs. In addition, there will be reduced diversity in the systems across the national partners. National partners that wish to implement a different systems stack will need to balance the additional costs of developing the broker interface for their systems against their perceived benefits. BT's approach to resolving this conflict is described in section 4.

3.3 National partner life cycle

BT's global systems strategy also has to take into account the national partner life cycle. This has been characterised into three phases:

- start-up phase,
- enterprise phase,
- operator phase.

The characteristics of these phases and their systems implications are given in Table 1.

BT has classified all its priority countries into gold, silver and bronze countries with gold signifying the aim to have a national partner with full second operator status offering domestic and global services, through to bronze countries which may only offer the Concert global portfolio through a distributorship. Most national partners will begin at the start-up phase but only the silver and gold countries can expect to progress to the enterprise and operator phases, as shown in Fig 2. The speed of this migration will vary from partner to partner and plays an important role in setting the systems strategy.

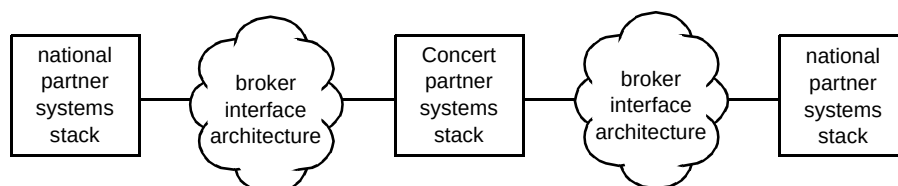


Fig 1 Schematic of brokering between national partners.

Table 1 Life cycle phases and their systems drivers.

Start-up phase - Initially address the corporate segment only for both global and local services. - It will be likely that with low start-up volumes, rudimentary, possibly manual integration of systems will be implemented rather than a potentially costly, integrated set of applications. This may even extend to reusing Concert systems on an agency basis. - Start-up costs need to be minimised. - There may be the need to incorporate legacy systems where appropriate.
Enterprise phase - Still focus on the corporate segment. - Increasing volumes will necessitate automation of most business processes.
Operator phase - National partner achieves second operator status and also addresses the local residential market. - Will need systems that cope with hundreds of thousands of residential customers. - Support for correspondent services, e.g. IDD.

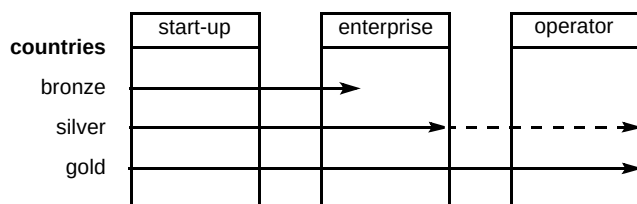


Fig 2 Classification of countries and mapping to national partner type.

3.4 Regulatory constraints

As market entry barriers fall across the globe, a set of regulatory barriers are emerging which will play a significant part in the systems of national partners. Regulation is becoming prevalent at three levels — national, regional and global.

Each and every country has today and will continue to have some sort of national regulatory body determining overall telecommunications policy. Even in the UK, the most liberalised market in the world, OFTEL continues to play a significant role. In Europe, EC Directives have already significantly influenced the telecommunications market by initially setting the telecommunications liberalisation date for 1998. At a global level, telecommunications is being seen as a key area for trans-border trade and will come under increasing scrutiny with GATT and the World Trade Organisation. Any tension between these three bodies will largely reflect national political forces. This three-level hierarchy is shown in Fig 3.

The significance of these bodies on systems strategy can be seen by analysing one example of a regional body — the open network provision (ONP) directive set out by the European Commission.

The aim of the EC Directive is to harmonise conditions of access to public networks for competitive service providers, though the ONP principles may take up to five years to be realised throughout Europe. The systems implications of the ONP directive have not been defined to date, but it is likely that the directive will include the specification of open interfaces for systems managing, for example, fault-handling and provisioning processes. Thus other operators or service providers can gain access to a network service from a network provider without the need to build infrastructure. BT's national partners could be in both or either of the network provider and service provider camps. Since the broker-interface architecture described earlier in this paper already focuses on open interfaces, this is a key 'future-proofing' element of BT's global systems strategy.

4. Global systems strategy

4.1 Systems scope

The scope of the global systems strategy has to cover both the traditional operational support systems (service and network management) and business support systems, as shown in the following list:

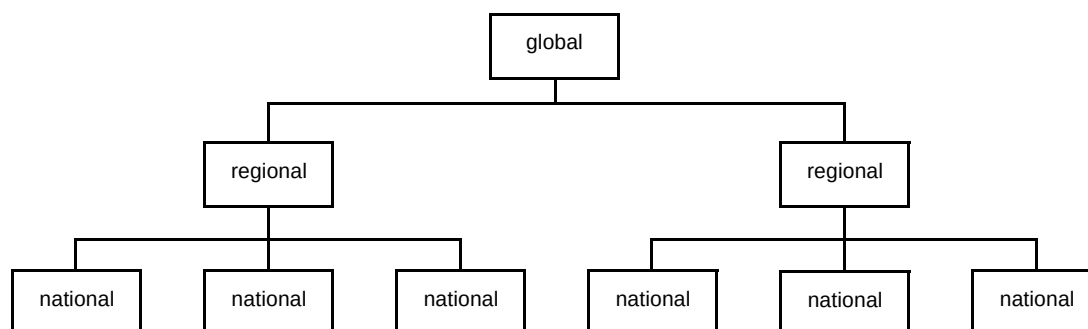


Fig 3 Hierarchy of regulatory bodies.

- customer handling,
- order handling,
- billing,
- business management (finance, logistics, personal),
- network inventory,
- network fault handling,
- network management,
- element management,
- sales and marketing.

BT and MCI are jointly developing components of the national partner systems stack to support the functions above, thereby reducing costs to BT's and MCI's national partners.

4.2 Mapping of systems strategy on to national partner phases

The definition of the three phases described in section 3 stated that the start-up and enterprise phases were confined to offering only corporate services. The proposed systems solution for the start-up and enterprise phases is built around mainly third-party solutions, integrated with strategic BT systems.

The systems solution for the start-up phase is characterised by few systems with limited integration, scaled for small volumes of customers and transactions. The start-up solution is a simplified design of the enterprise solution, where a number of the systems solutions are substituted by manual processes, and integration between the systems that are available may also be manual.

The migration to the enterprise phase is characterised by an integrated set of scalable core systems offering high functionality. The enterprise systems stack will also be flexible enough to cope with growth in business volume and product scope. The external systems interfaces are defined to ensure seamless interworking between national partners, based on the broker-interface specifications. However, there will be a degree of local autonomy as limited customisation will be allowed. A schematic of the start-up and enterprise stacks is shown in Fig 4.

The change from the enterprise phase to the operator phase is characterised by the introduction of services predicated on obtaining a second operator service and infrastructure licence. The second operator platform is not likely to be the enterprise platform scaled up to larger volumes as it will need to cater for millions of residential customers rather than thousands of corporate customers, and support a significant degree of churn. BT is proposing a modified systems stack for the operator phase which utilises those systems from the enterprise stack that need not scale up and a new set of mass-market systems for serving residential customers. These operator-phase systems are in

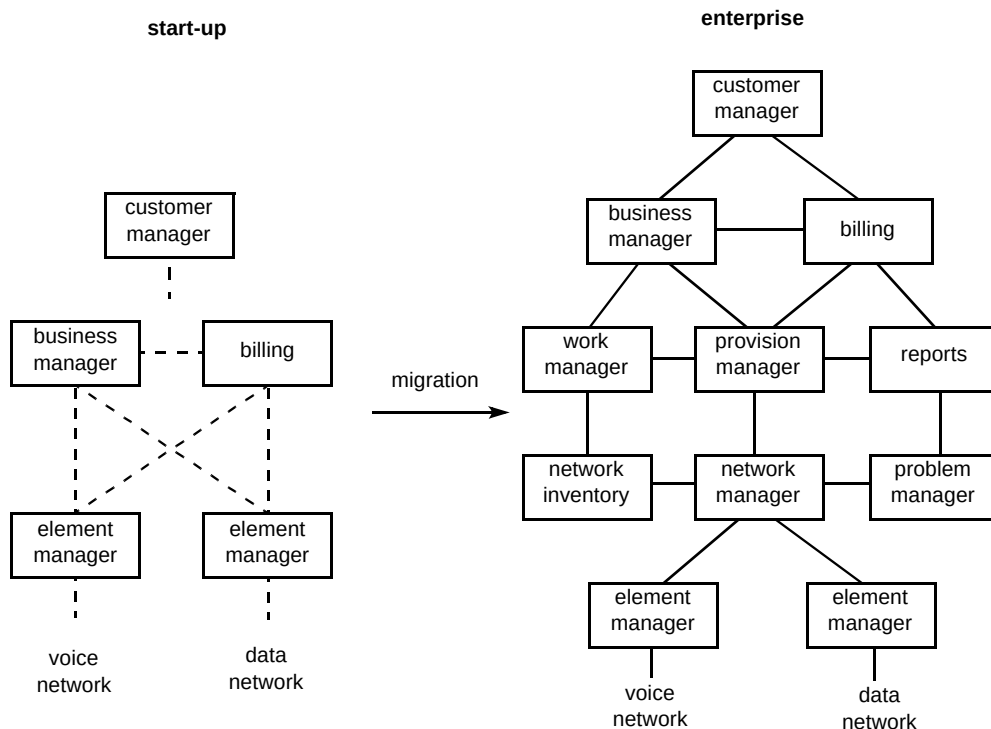


Fig 4 Systems strategy for start-up and enterprise phases.

general more expensive to install and maintain. Therefore it would not be economically viable to install these systems for the start-up and enterprise phases, and the systems strategy is currently to support the operator phase on separate systems stacks for the residential and corporate markets.

BT does, however, possess a host of large-scale systems in the UK which serve the residential market. These 'PTT-type' systems, in general, scale up to managing tens of millions of customers and are mostly too big for consideration. Figure 5 conveys these break points in systems scalability.

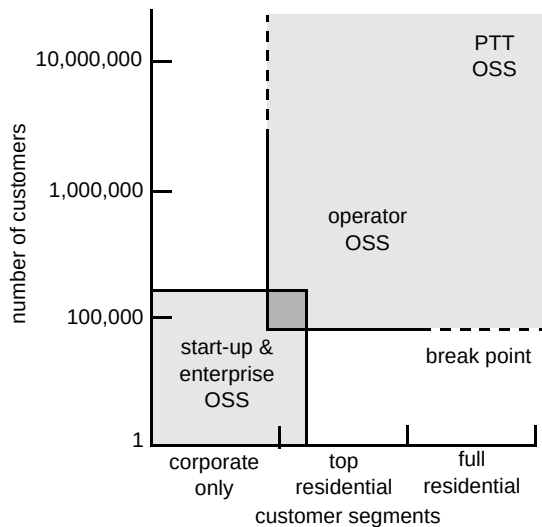


Fig 5 Systems scalability limits mapped on to customer segments.

In some liberalised markets, mobile licences may be offered to a second operator in addition to a fixed operator's licence. For these cases, the issue of fixed/mobile convergence needs to be considered, to ensure systems consolidation, wherever possible.

Fixed/mobile convergence (FMC) is a term that is often misquoted and misunderstood. The business drivers for FMC need to be carefully analysed before the implications for the system strategy can be discussed. To date, the most critical factor for operators seeking FMC has been the regulatory climate. If regulation dictates separate operations for fixed and mobile services, then separate operations centres and systems are inevitable. This section assumes that a national partner could offer both a fixed and mobile portfolio without any constraints. It is also worth noting that currently, no operator in the world manages a fixed and mobile operations service with a common set of systems. Hence the risk to a national partner is high if it aspires to only common systems for fixed and mobile services from start-up.

Three main business drivers have been identified for FMC:

- lower investment costs,
- lower operational costs,
- new products and feature sets.

The key to realising some or all of these drivers is the regulatory climate. A friendly regime will probably ensure that all these drivers are met to varying extents.

If regulation dictates separate operations for fixed and mobile services, then the lower investment and operational costs will be harder to materialise, though there may still be some opportunity for similarity of products and services feature sets.

5. Conclusions

The globalisation of BT via the formation of various national partner ventures produces a set of complex and rapidly evolving requirements on OSS. The need for a lucid systems strategy for national partners has been described which specifies the interface architecture to enable global interworking. The challenge now facing BT and its national partners is to continue joint development of the preferred systems stack progressively introducing the broker-interface architecture at all levels, for all partners in order to realise the cost savings and economies of scale across the globe.



Hari Sundaresan received a BA degree in Natural Sciences (Physics) from Cambridge University in 1987. After graduating he joined BT Laboratories in the optical devices division where he worked on tuneable semiconductor lasers. He was awarded a PhD for this work from Sheffield University in 1992.

He then moved to the Group Architect's Office where he worked on a number of strategic projects ranging from personnel profiling for BT to analysing BT's position in the systems integration market

Since 1993, he has been in the operational support systems sector where he led a team defining BT's global systems strategy. He has also been BT's systems advisor to a number of European joint ventures. He has recently become Data OSS Programme Manager, and is currently part of BT's 1996 Telecomms Masters Programme.